

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0

- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

Secure and Efficient Autonomous Systems Lab

Undergraduate Research Assistant

May 2026 – Present

East Lansing, MI

- Stress-tested the security of an LLM-controlled robot, targeting the cheap per-step gate that screens its motion plans as the exact seam an attacker slips a harmful plan through, one innocent fragment at a time.
- Reproduced a published adversarial-robotics attack loop (attacker, target, judge, syntax-checker) as the baseline, then drove the decomposition attack to an 80% success rate against the per-step triage gate.

MindMosaic

Software Engineering Intern

Aug 2025 – May 2026

East Lansing, MI

- Built and shipped an interactive knowledge-graph platform that keeps dense, interconnected content navigable as it grows, delivering a production MVP to real users on a polyglot Java Spring Boot, Neo4j, and PostgreSQL stack.
- Migrated hot read paths off recursive SQL self-joins onto a Neo4j graph layer so relationships stay first-class, cutting query latency 48% (850ms to 440ms) and making multi-hop traversals 3x faster than the equivalent SQL.
- Designed a dual-database layer that keeps transactional records in PostgreSQL while serving relationship-heavy reads from Neo4j, routing each query class to the store that answers it fastest instead of forcing one engine to do both jobs.

PROJECTS

Dadei | *Python, FastAPI, Redis Streams, pgvector, React, Docker*

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Routed all model-proposed side effects through one approval chokepoint with a cancelable countdown, giving every irreversible action one auditable seam between the LLM and the outside world.
- Engineered a no-merge-first speaker-ID model using an EMA voice centroid plus a bounded prototype bank with floor-and-margin gating, abstaining into a new identity on ambiguity, attributing 96% of utterances to the correct speaker.
- Replaced single-signal context lookup with hybrid retrieval over pgvector that fuses vector similarity, lexical overlap, recency decay, and participant overlap, grounding each proposal under a fixed token budget.
- Decoupled background workers from the live FastAPI service behind a Redis Streams event bus with consumer groups, dead-letter routing after 5 retries, and stale-claim recovery, giving at-least-once delivery that pub/sub dropped.

fliks | *Go, PostgreSQL, Object Storage, ML Rating Service*

github.com/fliks-gg

Multi-game platform that crowd-validates gameplay skill and mints shareable certification cards

- Built a self-hosted video transcoding pipeline with a concurrent Go worker that pulls upload jobs, transcodes, and writes processed video back to object storage, owning the media path end to end instead of renting a managed service.
- Architected an ML rating service that grades each clip independently and shows its verdict beside the crowd's, with those crowdsourced ratings serving as the labeled data the classifier trains on.

WizViz | *Python, MediaPipe, OpenCV*

devpost.com/software/wizviz

Webcam fighting game driven by real-time body-gesture recognition, winner of the Interactive Media track at SpartaHack X

- Classified five distinct combat gestures from body landmarks with a scale-invariant geometric model normalized by torso length, making detection work across players and camera distances without any labeled training data.

TECHNICAL SKILLS

Languages Python, Java, Go, JavaScript/TypeScript, C++, SQL

AI / ML Large Language Models, Machine Learning, Agentic Workflows, OpenCV

Databases PostgreSQL, Neo4j, Redis, pgvector

Cloud & Infrastructure AWS (EC2, S3, RDS), Docker

Tools Git